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IMPROVING KITCHEN PREP TIME PREDICTION

PS 1 | 3Members | Scalable | Signal-First

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Signal Engineering | System Design | Merchant Workflow

PRESENTATION AGENDA

What we will cover in the next 15 slides

01 **Problem Understanding**
The KPT chain & why accuracy matters

02 **Why Signal Is Noisy**
5 root causes of FOR button bias

03 **Signal Improvements**
4 high-impact signal-level solutions

04 **Kitchen Rush Score (RLRS)**
Estimating the invisible kitchen load

05 **Smart App Redesign**
4-stage workflow replacing single button

06 **System Architecture**
5-layer signal-to-prediction pipeline

07 **Team & Scalability**
Roles, phases, and 300K rollout plan

08 **Impact & Metrics**
Simulated gains and success criteria

The KPT Problem

Kitchen Prep Time is the foundation of every ETA Zomato shows customers

300,000+ Restaurant Partners

Even a 1-minute KPT error across 300K restaurants causes millions in idle rider costs, cancelled orders, and lost customer trust every day.

1. Order Placed
Customer sees ETA

2. DP Dispatched
Rider routed to restaurant

3. FOR Marked
Core problem: noisy signal

4. DP Picks Up
Idle time is early

5. Delivered
ETA error = cancellations

Why Current KPT Signal Is Noisy

5 root causes degrading model accuracy

FOR Button Misuse

Merchants press ready only when the rider walks in — not when the food is actually done.

Hidden Kitchen Load

Dine-in, Swiggy & walk-in orders are invisible — rush is severely underestimated.

DP Behaviour Noise

Old KPT = accept - pickup timestamp. DP detours polluted the restaurant signal.

Manual Inconsistency

Different staff tap FOR at different workflow stages — creating high label variance.

Single-Button Ambiguity

One button gives zero pipeline visibility: started? cooked? packed? waiting?

Our Solution: Signal-First Approach

Fix what goes into the model — not just the model itself

Noisy FOR Signal → GPS Correction → 4-Stage Workflow → Rush Score (RLRS) → Clean Prediction

PROBLEM	SECTION-A	SECTION-B	SECTION-C	OUTCOME
Noisy FOR Signal	GPS Correction	4-Stage Workflow	Rush Score (RLRS)	Clean Prediction

9%
9% gain from FOR button (Zomato data)

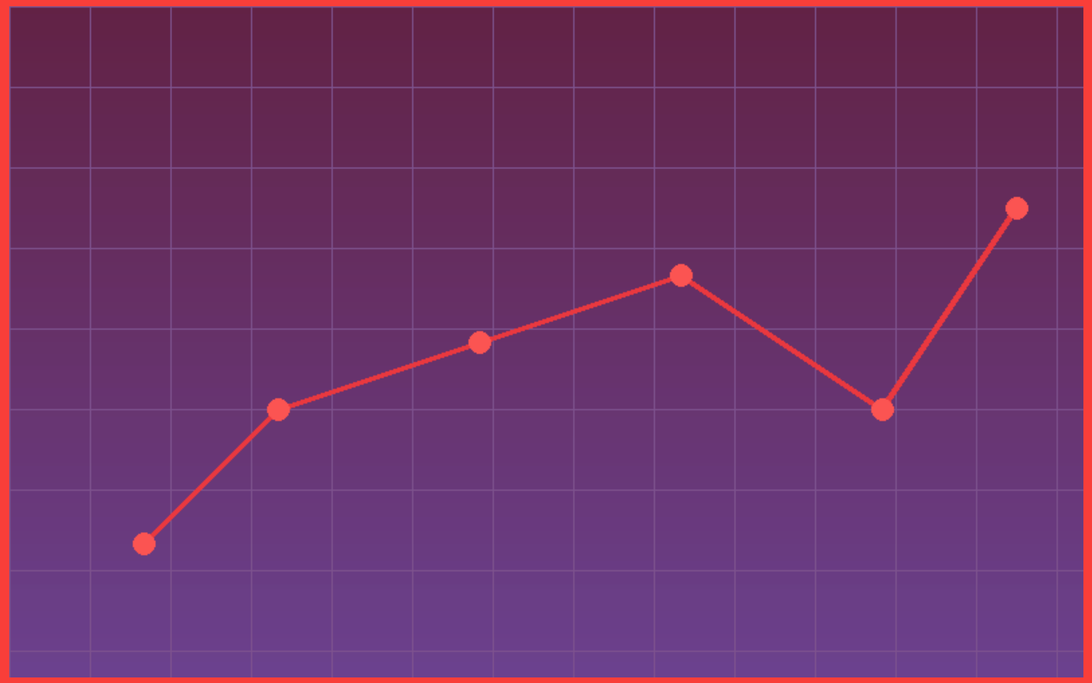
4.13 min
Currbaseline ent MAE

300K
Restaurant partners

~25%
Target accuracy improvement

Solution A

GPS-Based True Ready Time



Zero hardware needed.
Pure backend signal correction.

01

Rider GPS Geofence

Track when rider enters 50m radius of restaurant — establishes true DP arrival time

02

Wait Duration Measure

High wait after arrival = FOR was pressed late. Zomato auto-detects biased labels.

03

Pickup Scan Timestamp

QR or barcode scan at handoff = ground-truth upper bound for food-ready time.

04

Retrospective Label Fix

Use corrected timestamps to relabel historical data and retrain the KPT model.

Solution B: 4-Stage Smart FOR Workflow

Replace the single button with a granular kitchen progress signal — highest-impact change

1

ORDER STARTED

Chef acknowledged — establishes true cook-start time

2

COOKING DONE

All items cooked, moving to packing station

3

ORDER PACKED

Items packed, quality check in progress

4

READY FOR PICKUP`

Bag at the counter — rider can collect immediately

THANK YOU